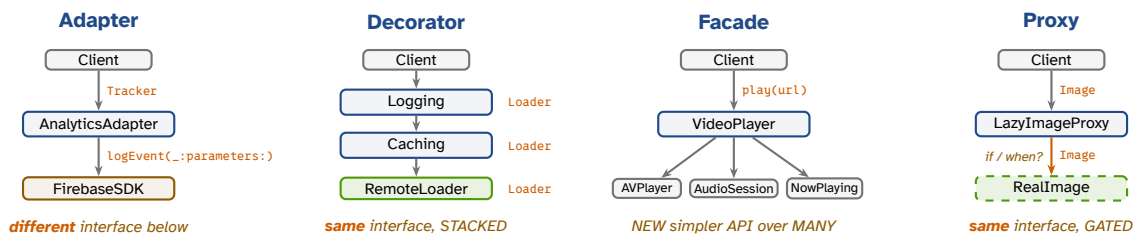


In one line: Creational patterns decide *how an object comes to exist* (Singleton, Factory Method, Abstract Factory, Builder). The four “wrappers” are told apart by **intent**, not shape: Adapter **changes** the interface, Decorator **keeps** it and **adds behaviour**, Facade **simplifies** a whole subsystem, Proxy **keeps** it and **controls access**.



Pattern	Interface	Intent	Wraps	iOS example
Adapter	changes to one a client already expects	compatibility	ONE adaptee; adds no features	extension SDK: MyProtocol; callback API → async via withCheckedContinuation
Decorator	same	augmentation	ONE, handed in by client; stackable	Logging(Caching(Remote())); SwiftUI ViewModifier, .padding().border()
Facade	new, simpler (invented)	simplification	a SUBSYSTEM of many classes	VideoPlayer over AVFoundation; URLSession over the network stack
Proxy	same	access control	ONE subject, often created/owned by the proxy	lazy var; NSXPCConnection.remoteObjectProxy; auth-gated client; image cache

Decide in two questions. (1) Same interface as the wrapped object? *No* → dictated by an existing client? **Adapter** · invented to hide many classes? **Facade**. **(2) Same** → *adds behaviour*, stackable? **Decorator** · decides *whether / when / how* you reach the real object? **Proxy**.

Proxy — “Very Private Rooms Cache” Surrogate with the subject's interface that **controls access**. Virtual: defer expensive creation (lazy var = built-in virtual proxy; not thread-safe) · Protection: check auth, then forward · Remote: local stand-in for another process/server (XPC, API stubs) · Caching / smart reference: memoise, ref-count.

Decorator Holds one object of the **same protocol**, forwards every call, adds behaviour before/after; wrappers **stack**, order chosen at runtime. **Beats subclassing**: N optional features → 2^N subclasses (log+cache+retry = 8); decorators = N types, composed (OCP). Cost: forwards the whole interface (keep it small, ISP). SwiftUI: `.padding().background()` ≠ `.background().padding()`.

Facade vs Adapter Facade does *not* lock the subsystem away (you may bypass it) and is one-way — the subsystem does not know it (a Mediator's peers talk back). Adapter = *one* type made to fit an *existing* interface.

Singleton Purpose: exactly one instance + global access point. `static let shared = X()` + `private init()`: static stored properties are initialised **lazily on first access, once, thread-safely** (swift_once). Only *creation* is safe — mutable state still needs an actor/lock.

Downsides: global mutable state · hidden dependencies (absent from every init) · **hard to test**: can't swap a fake, state leaks between tests, order-dependent · tight coupling (DIP) · app-long lifetime · SRP.

Fix: depend on a **protocol**, inject it; `.shared` only as default: `init(api: API = LiveAPI.shared)`. `URLSession.shared/FileManager.default` are *shared instances*, not enforced singletons.

Remember Adapter changes · Decorator decorates · Facade fronts many · Proxy protects. Hooks: plug adapter · gift-wrap layers · building front · bouncer. **Factory: Method = one, Abstract = family.**

Factory — three different things Simple factory (not GoF): a function with a switch returning a concrete type.

Factory Method: base class declares a creation method, a **subclass overrides it to decide ONE product**; base code calls it. Inheritance. iOS: override class var `layerClass` to pick a UIView's CALayer.

Abstract Factory: an **object** (protocol) with **several** make... methods for a **FAMILY** that must match: ThemeFactory (makeButton, makeLabel) as Light/Dark; Live vs Mock services. Composition: passed in.

Builder Solves **telescoping initialisers** (`init(a)`, `init(a,b)`, `init(a,b,c,...)`). Builds **one product step by step**; setters **return Self** for chaining; `build()` validates. **Director** = holds a fixed recipe of steps. **Swift: default + named params usually replace it**. URLComponents is a value-type builder; @resultBuilder is a compile-time DSL, related in spirit only.

Traps

! **Proxy ≠ Decorator**: Proxy *controls access*, one, often creates the subject; Decorator *adds behaviour* to what you hand it, stacked.

! **Facade ≠ Adapter**: invents a simpler API over many vs fits one to an existing API.

! A function returning an object is a *simple factory*, not Factory Method.

! “static let makes it thread-safe” — only its initialisation.